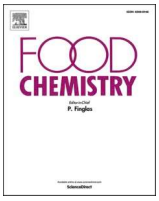
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Volatilomics and Lipidomics revealed flavoring mechanism in baijiu brewed from diverse Sorghum varieties

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ABSTRACT

Sorghum is crucial in Baijiu brewing. Previous studies have shown that glutinous sorghum (LS) and non-glutinous sorghum (FS) significantly affect Zaopei's lipid metabolism and lipids affect flavor perception. This study aimed to explore the quality differences in Baijiu brewed with different sorghum varieties and the effects of lipids. Sensory evaluation, volatilomics, and lipidomics were used to analyze differences in flavors and lipids. The results showed that significant differences in flavor existed between the two types of Baijiu. The aged flavor of LS was more pronounced and its mouthfeel was smoother. However, the FS showed a more intense aroma release. In addition, FA 18:1;O3 (fatty acyls), acetaldehyde, and ethyl caprylate contributed to the smoothness and softness of Baijiu. These results suggest that the lipid composition of Baijiu affects its aroma and taste, thus providing a reference basis for selecting suitable raw sorghum materials to enhance the quality of Baijiu.

1. Introduction

Baijiu is a traditional distilled spirit alcohol produced by the fermentation and distillation of grains. It has a long history of importance in the global alcohol market (Wu et al., 2024). The production of five-grain strong-flavor Baijiu (SFB) through a mud cellar solid-state fermentation process in an open environment is a more complex process in which multiple microorganisms and enzyme systems work together to give Baijiu a mysterious and complex flavor and sensory properties (Li, Zhang, et al., 2024). SFB has the largest consumption share of the 12 aroma types of Baijiu in China's aroma category, with the industry market size growing at a rate of over 10 % per year, and it is expected that the annual output value will exceed 300 billion RMB by 2025 (https://www.chinabgao.com/freereport/92387.html). Sorghum is primarily used as a raw material in traditional SFB brewing processes. Wheat, rice, glutinous rice, corn, and other grains are mixed and brewed in different proportions to improve the quality of Baijiu.

Sorghum is the fifth most prominent grain worldwide and the most widely used raw grain for producing high-quality Baijiu (Wu et al., 2017). Alcoholic distillation and steaming of fresh sorghum co-occur, and the aroma compounds produced during the steaming process are injected directly into freshly produced Baijiu(Jin et al., 2017). The aroma compounds of steamed sorghum considerably influence the final flavor quality of Baijiu (Fan et al., 2021), and differences in sorghum varieties also significantly affect the quality of Baijiu (Liu, Tang, et al., 2023; Liu, Yang, et al., 2024). The sorghum varieties used in Baijiu production are classified into glutinous and non-glutinous types based on their proportions of amylopectin and amylose (Chen et al., 2019; Liu,

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Tang, et al., 2023). Glutinous sorghum (GS) has higher fat, tannin, protein, and amylopectin content than non-glutinous sorghum (NGS), and southern sorghum has better starch stability and is more suitable for brewing (Tian et al., 2017; Yuan et al., 2011). In addition, previous non-targeted metabolomic studies by our group revealed significant differences in the lipids and amino acids of the two types of sorghum and its Zaopei (Li, Ma, et al., 2024). Different sorghum varieties significantly impact Baijiu yield, flavor substance type, and content owing to differences in their physicochemical properties, such as tannin and lipids. Currently, studies on the brewing characteristics of sorghum have focused on the physicochemical properties and differences in the sorghum itself, including starch structure and protein content (Mawouma et al., 2022; Xu et al., 2021). Several studies have investigated the impact of volatile organic compounds (VOCs) in Baijiu and the microbial composition on fermentation (Liu, Li, et al., 2023; Liu, Tang, et al., 2023).

SFB is one of the most ester-rich distilled spirit alcohols (Lin et al., 2023; Muñoz-Redondo et al., 2023; Xu et al., 2022). Current research on the esters of Baijiu focuses on VOCs, such as short- and medium-chain fatty acid compositions represented by the four major acids and esters. However, few studies have been conducted on non-volatile compounds (Jia et al., 2020; Xu et al., 2022). Lipids can change the flavor perception by interacting with other compounds in Baijiu (Liu, Lin, et al., 2024; Saad et al., 2021; Wu, Chen, Huang, et al., 2023; Wu, Chen, Sun, et al., 2023). Recent studies have found that long-chain fatty acid ethyl esters (LCFAEEs) significantly correlate with the sensory experience of Baijiu, including the initial taste and aroma sensation in the mouth, as well as the evaluation of taste after drinking (Sun et al., 2023; Wu, Chen, Huang, et al., 2023). Medium- and long-chain fatty acids alter Baijiu's intermolecular interactions between volatiles, EtOH, and water molecules and have positive effects in reducing the Baijiu irritating odor (Liu, Lin, et al., 2024). Simultaneously, our preliminary in-cellar test revealed the differential enrichment of Zaopei's differentially expressed substances from various sorghum raw materials in the triglyceride metabolism pathway (Zhu et al., 2025). Although the type and content of lipids entering Baijiu after distillation are reduced (Jia et al., 2021; Jia et al., 2023; Jiang et al., 2021; X. Zhang et al., 2022), recent studies have found that the lipid components in distilled Baijiu significantly affect flavor rendering. Current research on alcohol-related lipidomics has only focused on non-distilled spirits such as wine (Casalta et al., 2019; Gómez et al., 2004; Phan et al., 2022; Phan & Tomasino, 2021; Tumanov et al., 2015), and there is a lack of research on Baijiu lipids.

Therefore, this study focused on the lipidomics of Baijiu. Using different sorghum varieties (GS and NGS) as research subjects, production-brewing experiments were conducted to explore the impact of different raw materials in brewing Baijiu on the quality of Baijiu. The sensory properties of the two Baijiu samples were initially obtained using flavor wheel sensory evaluation, *E*-nose, and E-tongue, and the different VOCs of the two Baijiu samples were further identified by HS-SPME-GC–MS. Finally, the Baijiu lipid profiles were obtained using UHPLC-TOF-MS/MS. Moreover, the sensory differences in Baijiu, odor differences in VOCs, and composition and content of lipids will be systematically compared, and the relationship between the sensory characteristics, VOCs, and lipids in Baijiu will be discussed. To provide a reference basis for selecting raw materials for Baijiu brewing and improving the quality of SFB.

2. Materials and methods

2.1. Baijiu sample

The Baijiu samples were obtained from Sichuan Yibin Hengshengfu Liquor Industry Group Co., Ltd. A high-quality cellar line was used to perform a continuous comparative in-cellar production test. During the test, several rounds of adjustments were made to the ratio, distillation time, and cell temperature until the ideal parameters were determined. These parameters were uniformly maintained during the subsequent brewing process for the control test. The "running cellar cycle" process was adopted, and after several cycles, the same point was chosen to collect Zaopei for Baijiu distillation. Sixteen Baijiu samples were collected in this study, with eight brewed from GS ((sample LS) and eight brewed from NGS (sample FS). The samples were brewed using the same technology in different pits within the same factory area during the same period and were aged for three years.

2.2. Sensory evaluation of baijiu

Candidates for sensory evaluation were recruited from students at the School of Bioengineering, Sichuan University of Science and Engineering, with preferences given to those with sensory analysis experience and familiarity with Baijiu. In the sensory analysis laboratory, the volunteers were subjected to basic olfactory, aroma matching, sequencing, and descriptive ability tests. They were initially selected to form a group of candidate sniffers based on a 70 % accuracy rate in the previous tests. They underwent a two-week training program (lasting 60 mins per day), which included training in basic knowledge of SFB, training in the basic ability to review Baijiu, and identification and description of SFB sensory characteristics. The evaluators sniffed and memorized the aroma reference substances five times a week for 30 min each time. Finally, the assessor had to achieve an accuracy of 80 % for blind tasting before passing the test. Simultaneously, a Baijiu sample was selected each week, the evaluators were asked to describe the perceived aroma of this sample, and the results were discussed and shared. Twelve well-performing evaluators (six females and six males) were selected as part of the evaluation panel. Baijiu samples were prepared in triplicate. Samples (15 mL) were served in standard Baijiu tasting glasses, randomly numbered, and scored with reference to the flavor wheel, and sensory data were collected manually. All human sensory experimental protocols complied with the relevant laws and were approved by the University Scientific Research Ethics Committee of Sichuan University of Science and Engineering (Ethical Review Number (2025LS101)).

The evaluation room was kept clean and orderly, with a temperature maintained at 20 °C. Baijiu samples were numbered and grouped accordingly. Typically, Baijiu evaluations were conducted either from 9 to 11 o'clock or from 15 to 17 o'clock. The sensory assessment was based on Liu Ying's flavor wheel evaluation method (Liu et al., 2022), using a nine-point scale to score the aroma, fragrance, taste, mouth feel, and style, and the quantitative expression of the characteristics referenced in Supplementary Table 1.

2.3. Odor analysis by GC-E-nose

The methods described to (He et al., 2021) were used for gas chromatography (GC) E-nose analysis. Heracles NEO (French Alpha MOS) was equipped with two different polarized columns; the columns consisted of MXT-5 and MXT-1701 (20 m, 0.18 mm ID, 0.4 μm) plus two FID detectors. Samples were analyzed by GC-E-Nose and diluted 1:3 in 15 mL headspace vials. Hydrogen was used as a carrier gas, and all samples were run three times in headspace vials. The average of the three replicate determinations was taken as the value of each response for analysis in the post-processing data.

Instrument parameters: Inlet volume 1000 μL, incubation temperature 50 °C, incubation time 20 min, trapping temperature 40 °C, injection rate 125 μL, syringe temperature 60 °C; inlet temperature 200 °C, inlet pressure 10 kPa, outlet 30 mL/min, trap original temperature 40 °C, pre-column pressure 54 kPa, shunt 10 mL/min, constant current 1.0 mL/min, trapping time 18 s; cleaning time 90 s. Temperature and pressure program: initial temperature 40 °C (5 s), stirring speed 500 rpm, and then increase by 4 °C/s to 240 °C (30 s).

2.4. Determination of taste by E-tongue

Methods referring to the study of (Wang et al., 2024), briefly, before testing the sample, the ASTREE electronic tongue (French Alpha MOS) was pre-balanced and calibrated with distilled water and 0.01 mol/L hydrochloric acid solution. To ensure the stability and reliability of data collection, the experiment is always maintained at 20 °C. The Baijiu sample, was to 1:3, add distilled water 4 times for dilution, and the diluted Baijiu sample was transferred to a 150 mL electronic tongue beaker. The data acquisition time is 120 s, the stirring speed is 1 r/s, the acquisition period is 1 s, and the cleaning time is 120 s. The average of three replicate measurements was used as the response value for each taste, and the data were used for post-processing analysis. The sensors and taste information for the ASTREE *E*-tongue are listed in Table 1.

2.5. Determination of VOCs by HS-SPME-GC–MS

Methods referring to the study of (Zhang, Shi, et al., 2022), briefly, a total of 5.0 mL of the sample was transferred into a 20-mL headspace vial. Subsequently, 15 μL of 2-octanol solution was added as an internal standard. After the SPME fiber was inserted into the sample for adsorption, the odor was analyzed (Agilent 7890 B GC and 5977 A MSD).

The flavor components of Baijiu were qualitatively analyzed by comparing the automatically identified chromatographic peaks with the National Institute of Standards and Technology (NIST) 12 mass spectrometry library through retrieval and combination with the retention index (RI). The internal standard method was used for the quantitative analysis.

2.6. Odor activity value (OAV) calculation

OAV determines whether a compound generates an aroma. Compounds with OAV ≥ 1 contribute to the overall olfactory characteristics of the substance.

$OAV = C/OT$.

Here, C: the mass concentration (μg/L) of the compound calculated according to the internal standard method, and OT: the odor threshold of the aroma compound referred to in the literature, referring to the threshold of the compound in 53 %, Vol ethanol/aqueous solution.

2.7. Lipid extraction of baijiu

In a previous study of (Phan & Tomasino, 2021) briefly, the Baijiu samples were concentrated by rotary evaporation. Trichloromethane (5 mL) and methanol (10 mL) containing 0.005 % (*w/v*) BHT were added to 4 mL of the concentrated Baijiu sample. The solution was mixed by shaking the table at room temperature at 165 rpm for 1 h. Five milliliters of trichloromethane and 5 mL of ultrapure water with 0.8 % KCl added to the mixture, and the final retention of trichloromethane: methanol: concentrated Baijiu is 1:1:0.9 v /v. The mixture was shaken at 165 rpm for 15 min and then transferred to a 50 mL centrifuge tube. The organic layer was collected in a 20-mL amber vial and dried under nitrogen flow, and the drying results were divided based into the total lipid content.

Table 1
ASTREE E-tongue sensors and taste information.

code	sensor name	taste information
1	AHS	sour
2	PKS	universal
3	CTS	salty
4	NMS	umami
5	CPS	universal
6	ANS	sweet
7	SCS	bitter

2.8. UPLC-TOF-MS/MS analysis

The instrumental method was as described by (Phan & Tomasino, 2021).

2.9. Data processing, statistics, and visualization

Data analysis was performed using SPSS version 26.0. The significance of the data was statistically analyzed using one-way analysis of variance (ANOVA). Principal component analysis (PCA) and orthogonal partial least-squares discriminant analysis (OPLS-DA) of the data were performed using MetaboAnalyst 5.0 (http://www.metaboanalyst.ca/). Significant metabolites were identified according to the variables that met the variable importance in projection (VIP) > 1 criterion. Spearman's correlation analysis was used to assess the relationships between the sensory characteristics, VOCs, and lipid profiles of Baijiu.

3. Results and discussion

3.1. Sensory evaluation

Sensory evaluation of SFB from different sorghum cultivars PCA is an unsupervised statistical analysis method that can summarize differences in composition and assess the regularity and variability of the samples (Li et al., 2019). The PCA results of the sensory evaluation showed that the contribution of PC1 was 78.8 % and that of PC2 was 9.8 %, with a cumulative contribution of 88.6 % (Fig. 1a), and the sensory evaluations of the two samples were differentiated from each other. It is speculated that the reasons for these differences are related to the influence of the characteristics of the raw materials and the microbial structure (Liu, Tang, et al., 2023).

The results of the sensory scores of the two samples are shown in Table 2. The sensory descriptive analysis of the two samples showed obvious differences. In terms of aroma, the aged fragrance and aroma sensation of Baijiu LS were better than those of FS, which made the aged fragrance more obvious, elegant, harmonized, and comfortable. The aroma of the raw materials and fermentation of FS were significantly better than those of LS Baijiu. The sense of the FS was also stronger. In terms of taste and mouthfeel, LS is more pronounced in terms of sweet, sour, and umami and is more coordinated and soft. There was no significant difference between the styles of the two samples; however, there was a substantial difference in aroma and taste.

To further clarify the sensory discrepancy indexes of different Baijiu samples, the sensory indices with significant differences were analyzed according to the OPLS-DA model, with variable importance projection (VIP) > 1 as the screening criterion. The results are shown in Fig. 1c; 21 key sensory difference indices were screened. Ranked in order of VIP value are sweet, alcoholic aroma, alcoholic aged, fragrant, nutty aroma, comfort, sour, Zaopei aroma, and so on.

The heatmap results (Fig. 1d) showed that LS had higher levels of umami, sweet, sour (taste), comfort, elegance (aroma sensation), fullness, softness, purity (mouth feel), and alcohol-aged fragrant (aged fragrant). In comparison, FS had higher levels of Zaopei aroma, pit aroma, alcoholic aroma, nutty aroma, acid aroma, woody (aroma of fermentation), wheat aroma, sorghum aroma, Jiuqu aroma, corn aroma, bran aroma, and grain aroma (aroma of raw materials). Overall, the aroma of the raw materials and fermentation of FS was superior to that of LS, and LS's aged fragrant aroma, aroma sensation, taste, and mouthfeel were superior to those of FS.

3.2. The result of odor analysis by GC-E-nose

PCA concentrates the odor information collected by the E-nose sensor to extract the main features. The samples were analyzed using plane coordinates, and the results are shown in Fig. 2a. The two samples differentiated, indicating some differences between the two odors: PC1

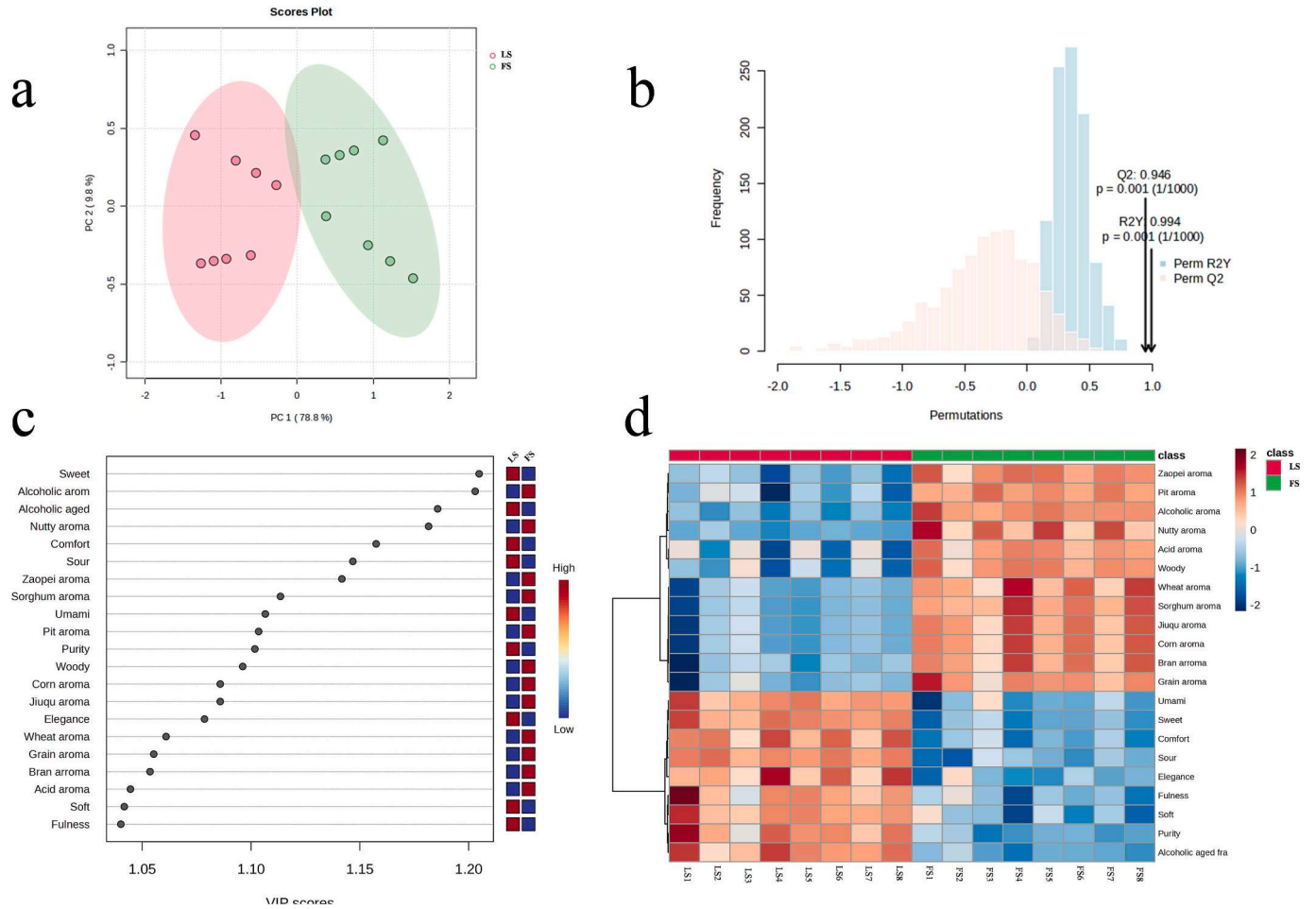


Fig. 1. Results of the sensory evaluation of Baijiu made from different sorghums. a). PCA of sensory evaluation results. b). Orthogonal Partial Least Squares-Discriminant Analyze (OPLS-DA) model permutation. c). 21 discrepant sensory characters (VIP>1) of different Baijiu. d). The heatmap of 21 sensory characters of different Baijiu.

was 98.6 %, PC2 was 1.2 %, and the cumulative contribution was 99.8 %. The two sorghum-brewed Baijiu samples were well differentiated. These results are consistent with the significant differences observed in the sensory evaluation of the aroma.

3.3. *The result of taste analysis by E-tongue*

Fig. 2b shows the E-tongue results for the different samples. PC1 was 63.9 %, PC2 was 20.8 %, and the cumulative value was 84.7 %. Clear differences in taste was observed between the two Baijiu samples.

The average of each the response values of each sensor was used to create a radar chart, where each branch represents one sensor, and the resulting closed curve is the digital information of the taste characteristics of Baijiu. The closure curves of the different Baijiu samples were not identical, indicating that differences existed in the taste information of the samples (Fig. 2c). The range of sensor response values for the Baijiu samples is 0–3500 (2.76–3133), with the lowest response value for the SCS sensor (<400) and the highest response value for the CPS sensor, followed by AHS (>2500), NMS (>2400), ANS (>2200), and PKS (>2000), and CTS (>500), respectively. The *E*-tongue results showed a significant difference between the AHS, ANS, NMS, and AHS sensor response values of the different samples, with the scores being LS > FS. These results are consistent with the sensory evaluation of Baijiu as sour, sweet, and umami.

3.4. ***Results of VOCs in different baijiu samples and OAV results***

Sensory evaluation results showed differences in the aroma and taste of the Baijiu samples, and GC–MS was used to explore the differences in the Baijiu flavors. The VOCs of Baijiu were qualitatively determined using GC–MS and quantitatively determined using an internal standard method, and a total of 69 VOCs were identified. There were 25 esters, 17 alcohols, nine acids, seven aldehydes, seven ketones, one nitrogenous compound, and three other compounds (Table 3). As shown in Supplementary Fig. 1a, LS and FS contained 66 and 69 VOCs, respectively. In LS, 23 esters, 16 alcohols, nine acids, seven aldehydes, seven ketones, one nitrogenous compound, and three other compounds were identified. FS identified 25 esters, 17 alcohols, nine acids, seven aldehydes, seven ketones, one nitrogenous compound, and three other compounds (Supplementary Fig. 1).

The species and proportion of VOCs in different Baijiu samples are shown in Supplementary Fig. 1b and c, which show that the content of esters in Baijiu was the highest, followed by acids and alcohols. The GC–MS results showed that all substances were detected in both Baijiu samples, except for ethyl phenylacetate, ethyl laurate, and glycerol. Among these, there were significant differences in the ethyl propanoate, ethyl isobutyrate, ethyl isovalerate, ethyl caprylate, ethyl phenylacetate, ethyl oleate, 3-furancarbinol, acetaldehyde, acetal, 2-heptanone, acetoin, and 1-hexanoic acid contents.

To further analyze the characteristic aromatic substances of the main flavor components in Baijiu, the OAV of the VOCs in SFB were calculated. The contribution of a single substance to the overall aroma of

Table 2
Sensory evaluation scores of different Baijiu samples (* P<0.05).

Class1	Class2	LS	FS
The aroma of raw materials	Grain aroma	6.95 ± 0.25	7.29 ± 0.21
	Sorghum aroma*	4.95 ± 0.25	5.70 ± 0.18
	Rice aroma*	5.09 ± 0.27	5.63 ± 0.26
	Wheat aroma*	4.95 ± 0.25	5.63 ± 0.26
	Corn aroma*	4.95 ± 0.25	5.55 ± 0.18
	Bran arroma*	5.02 ± 0.24	5.55 ± 0.18
	Jiuqu aroma*	4.95 ± 0.25	5.55 ± 0.18
Aroma of Fermentation	Alcoholic aroma*	7.36 ± 0.21	7.86 ± 0.18
	Pit aroma	7.47 ± 0.40	7.88 ± 0.13
	Zaopei aroma*	5.41 ± 0.23	5.75 ± 0.15
	Fruital	5.93 ± 0.27	5.98 ± 0.29
	Floral aroma	6.43 ± 0.27	6.05 ± 0.22
	Nutty aroma	5.73 ± 0.14	5.93 ± 0.20
	Woody	5.43 ± 0.27	5.75 ± 0.15
	Sweet aroma	7.09 ± 0.36	7.20 ± 0.09
	Acid aroma	5.48 ± 0.32	5.75 ± 0.15
Aged fragrant	Alcoholic aged fragrant*	6.93 ± 0.16	5.79 ± 0.17
	Pit aged fragrant	5.63 ± 0.10	4.90 ± 0.51
	Jiuqu aged fragrant*	5.18 ± 0.16	4.61 ± 0.21
	Old aged fragrant*	5.05 ± 0.09	4.16 ± 0.38
	Woody aged fragrant*	4.93 ± 0.16	4.16 ± 0.38
	Sauce aged fragrant*	4.93 ± 0.16	4.16 ± 0.38
	Oleic aged fragrant*	4.93 ± 0.16	4.16 ± 0.38
Aroma sensation	Intensity*	7.79 ± 0.31	6.88 ± 0.30
	Complexity	7.11 ± 0.44	6.20 ± 0.25
	Elegance*	6.13 ± 0.30	4.98 ± 0.24
	Comfort*	7.41 ± 0.26	6.13 ± 0.26
	Harmonize*	7.36 ± 0.21	6.32 ± 0.30
taste	Sweet*	7.29 ± 0.21	5.57 ± 0.27
	Sour*	6.66 ± 0.17	5.41 ± 0.24
	Bitter	6.54 ± 0.17	6.09 ± 0.38
	Umami*	5.61 ± 0.18	4.46 ± 0.31
mouthfeel	Soft*	7.41 ± 0.17	6.47 ± 0.33
	Fulness*	7.40 ± 0.29	6.47 ± 0.27
	Coordinate*	7.56 ± 0.41	6.59 ± 0.22
	Purity*	7.59 ± 0.22	6.66 ± 0.17
	Durability*	6.11 ± 0.22	5.48 ± 0.32
style	Typicality	7.59 ± 0.33	7.55 ± 0.22
	Obviousness	7.59 ± 0.33	7.55 ± 0.22

Baijiu can be assessed using OAV analysis based on the threshold value of each flavor substance. A total of 26 characteristic VOCs were identified (Table 4).

In the Baijiu samples, 26 VOCs with OAV > 1 were found, including 10 esters, six aldehydes, four acids, two alcohols, three ketones, and one nitrogenous compound, ethyl hexanoate, ethyl caprylate, ethyl valerate, isovaleraldehyde, and ethyl butyrate. There were significant differences between ethyl isobutyrate, ethyl isovalerate, ethyl caprylate, acetaldehyde, acetal, acetoin, 2-heptanone, and 1-hexanoic acid. The OAV results showed that the OAV values of most compounds in Baijiu FS were higher than those in Baijiu LS, which was speculated to be the reason for Baijiu FS being more obvious in fragrance and strong stimulation.

Esters play an important role in Baijiu and are the main factors in the formation of a strong aroma. Of all the flavor compounds, esters account for more than half of the total mass of flavor substances. Because most esters have a low odor threshold, they contribute significantly to odors (Qian et al., 2019). Esters are the most important class of aroma components in Baijiu. It is mainly through the biosynthesis of yeast or in the distillation and storage process of Baijiu esterification that primarily plays the role of presenting aroma and its content and proportion, which are key factors affecting the typical style and quality of SFB. Esters are the main aroma components of Baijiu; ethyl butyrate, ethyl isobutyrate, ethyl nonylate, ethyl isovalerate, and other ethyl ester compounds are mostly fruity, floral, and have unique aromas (Xu et al., 2017). Alcohols, aldehydes, and ketones mainly set off the ester aroma and harmonize the aroma and taste. In addition, many of these esters, along with acids, aldehydes, and alcohols, can influence the flavor of Baijiu (Wei et al., 2020).

The ester contents in samples LS and FS were 59.36 % and 61.00 %, respectively, and the variety of ester compounds was also the richest, with approximately 1/3 of the number of VOCs in the Baijiu samples. Ethyl hexanoate, ethyl acetate, ethyl lactate, and ethyl butyrate are the most representative esters in SFB, and their concentrations and ratios significantly enhance the flavor of Baijiu (Wei et al., 2020). There were no significant differences in the concentrations of the four typical esters found in the Baijiu samples, ethyl hexanoate, ethyl acetate, ethyl lactate, and ethyl butyrate, as well as the total esters (Supplementary Fig. 2a). The ethyl hexanoate content was higher in SFB, which has a lower aroma threshold and is a characteristic component of Baijiu (Chen et al., 2022; Wu et al., 2016; Xu et al., 2017). Ethyl acetate in the LS and FS accounted for 37.09 % and 37.05 % of the total ester content, respectively. Ethyl lactate is important for maintaining the integrity of Baijiu style. To preserve the strong aroma characteristics of Baijiu, the ethyl lactate content should be lower than that of ethyl hexanoate (Wang et al., 2022). The results of the GC–MS analysis supported this finding. Interestingly, ethyl caprylate content is correlated with Baijiu sweetness (Sun et al., 2021). Moreover, the OAV of Baijiu LS was higher in ethyl caprylate than that of Baijiu FS, which made Baijiu LS sweeter than Baijiu FS.

Acids are the essential aromatic compounds in Baijiu, and they contribute to the flavor and aroma of Baijiu, primarily by reducing off-flavors and bitterness. Low levels of acids produce a light aroma in Baijiu, whereas excessive acids can cause an off-flavor. Moderate amounts of acids act as buffers and slowly form esters with alcohol during storage. Acids can reduce the pungency and bitterness of Baijiu, increase its sweetness, and significantly affect its taste. The concentrations of 1-hexanoic acid, acetic acid, lactic acid, and butyric acid in SFB accounted for 94 % of the total acid content (Wu et al., 2022). In the

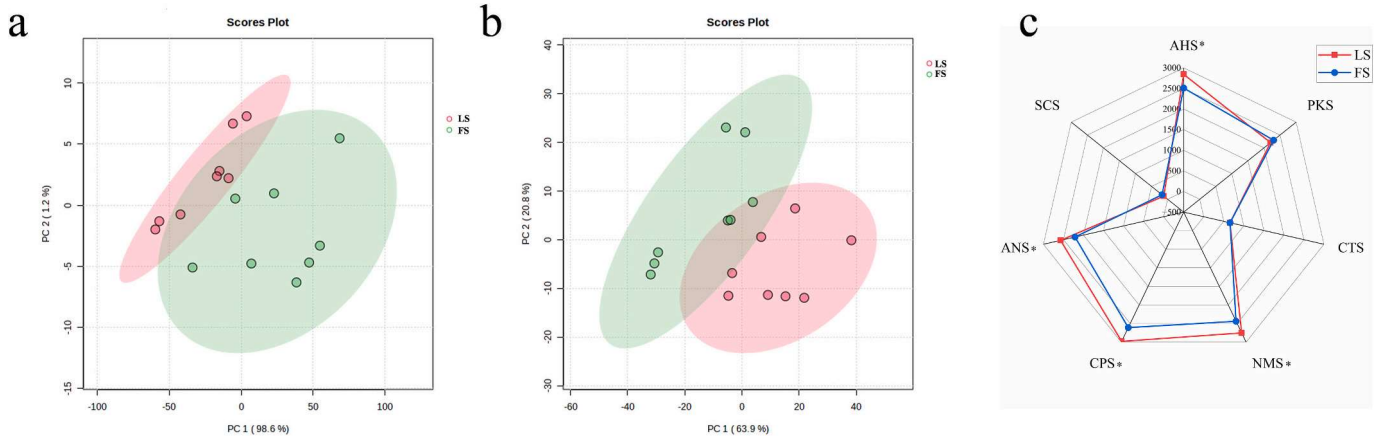


Fig. 2. Result of *E*-nose and E-tongue. a) The PCA of E-nose. b) The PCA of E-tongue. c) Radar determined taste results of different Baijiu.

GC–MS results, 1-hexanoic acid, acetic acid, and butyric acid of LS and FS accounted for 95.33 % and 95.22 % of the total acid content, respectively. Acetic acid is highly stimulating, imparting a pleasant aroma and sour flavor to Baijiu; however, excessive amounts can create a sharp taste. Butyric acid contributes a "Jiaoni" aroma and slight sweetness (Duan et al., 2015; Wu et al., 2022), and there is no significant difference between the content of acetic acid and butyric acid in the two samples (Supplementary Fig. 2b). 1-Hexanoic acid is one of the main components of the pit aroma (Wu, Zhu, et al., 2021). There was a notable difference in the concentration of 1-hexanoic acid in the GC–MS results, and a significant difference in 1-hexanoic acid was observed in the OAV results for Baijiu. FS was higher than that of LS, suggesting that FS had a more pungent pit aroma.

Alcohols are important for enhancing the sweetness and aroma of Baijiu, and are the primary source of esters. The most abundant alcohol in Baijiu is ethanol, while other alcohols include methanol, 1-propanol, 2-butanol, isobutanol, 1-butanol, isoamyl alcohol, 1-hexanol, pentanol, hexanol, heptanol, glycerol, 2,3-butanediol, and phenethyl alcohol (Chen et al., 2022). Alcohols containing more than three carbon atoms per molecule are classified as higher alcohols. Among the higher alcohols, isobutanol is known for its strong bitter taste, whereas 1-butanol and 1-propanol exhibit slight bitterness, and isoamyl alcohol has a combination of slight sweetness and bitterness (Fu et al., 2021). To achieve the best taste, Baijiu must contain an appropriate amount of alcohol(Han et al., 2017; Sun et al., 2022). There were no significant differences in the isobutanol, 1-butanol, and isoamyl alcohol contents found in the GC–MS results (Supplementary Fig. 2c), indicating that there was no significant difference in the bitterness of the two Baijiu samples. This result was in agreement with the results of the sensory evaluations and *E*-tongue. Isoamyl alcohol and isobutanol significantly affect the appearance and taste of Baijiu. Their content in high-quality SFB has a golden ratio of 3:1 (Xiao, 2011). In Baijiu LS and FS, the ratios of isoamyl alcohol to isobutanol were both 3.26:1, which is close to the golden ratio of SFB. He et al. (He et al., 2020) used GC × GC/TOFMS to comprehensively analyze SFB from different regions. They found that 1-butanol is the key flavor substance of SFB, and the results of this study indicated that 1-butanol OAV > 1 could be used as a differential substance to distinguish between the Baijiu samples.

The aldehydes in SFB primarily include formaldehyde, acetaldehyde, 2-furaldehyde, butyraldehyde, pentanal, and acetal. Most of these are produced by oxidation of the corresponding alcohols during fermentation (Fan et al., 2018; Sun et al., 2016). Acetaldehyde and acetals are important aromatic compounds in Baijiu, and the acetal formed by the condensation of acetaldehyde and ethanol gives Baijiu a pleasant aroma and taste (Wang, 2022). Acetaldehyde and acetals, as indicators of aging in Baijiu, showed a clear pattern of change, in which the content of acetaldehyde with the extension of storage time first increased and then decreased, and acetaldehyde with the storage time increased.

3.5. Differential analysis of VOCs in different baijiu samples

VOCs with significant differences were analyzed using the OPLS-DA model to further clarify the discrepant flavor compounds. The results are shown in Supplementary Fig. 3a. The LS and FS samples were positioned on the left and right sides of the horizontal coordinates, respectively, indicating that the overall flavors of the two samples were significantly different. The OPLS-DA model could differentiate the flavors of the original Baijiu with different raw brewing materials. Cross-validation through 1000 permutation tests confirmed the reliability of the OPLS-DA model with Q2 and R2Y values of 0.939 and 0.995, respectively (Supplementary Fig. 3b). This result matched the *E*-nose results for Baijiu aroma detection, indicating that there were differences in the VOCs in the Baijiu samples.

Twenty key discrepant flavor compounds were screened using variable importance for projection (VIP) > 1 as the screening criterion. The larger the VIP value, the greater the contribution of the variable in the discrimination process, and the more significant the difference between different samples, including six esters, six alcohols, four aldehydes, three ketones, and one pyrazine. The difference in flavor compound content may be an important reason for the difference in flavor between the LS and FS samples. The VIP values, ranked in order of VIP value were acetaldehyde, acetoin, acetal, ethyl propionate, ethyl caprylate, 2,3-butanediol, hexanal, isoamyl alcohol, isovaleraldehyde, and 2-heptanone. Pentanol, trimethyl-pyrazine, 2-butanone, 1-propanol, ethyl isobutyrate, ethyl phenylpropiolate, 1-hexanol, ethyl laurate, 1-butanol, and ethyl oleate. The heatmap of the 20 discrepant VOCs in the samples from the two groups are shown in supplementary Fig. 3d, and the content distribution of the discrepant compounds in the two groups was consistent with the analytical results of the VIP score plots.

The results showed that ethyl propanoate, ethyl caprylate, ethyl isobutyrate, ethyl phenylpropiolate, ethyl laurate, and ethyl oleate were the key discrepant substances between the LS and FS, whereas the GC–MS results showed significant differences in ethyl propanoate, ethyl isobutyrate, ethyl isovalerate, ethyl caprylate, ethyl phenylacetate, and ethyl oleate. The contents and compositions of medium- and long-chain esters in Baijiu were significantly different. Therefore, the lipidomics of Baijiu were further determined to obtain the lipid profile of Baijiu and to explore the lipid differences between the two types of Baijiu.

3.6. Lipidomics analysis of baijiu

The results were shown in supplementary Fig. 4a; the total lipid contents of samples LS and FS were 0.73 ± 0.11 and 0.62 ± 0.12 μg/mL, respectively, with no significant difference ($P = 0.0536$). Six categories

Table 3
Qualitative and quantitative results of volatile compounds in different Baijiu. (* $P<0.05$).

code	Compound	CAS Number	LS (μg/L)	FS (μg/L)
	Esters			
1	Ethyl formate	109–94-4	1655.80 ± 0.35	1720.01 ± 0.28
2	Ethyl acetate	141–78-6	177,484.06 ± 22.53	179,030.45 ± 20.75
3	Ethyl Propanoate*	105–37-3	1000.00 ± 0.04	743.75 ± 0.08
4	Ethyl isobutyrate*	97–62-1	441.81 ± 0.04	524.64 ± 0.05
5	Ethyl butyrate	105–54-4	10,833.05 ± 1.46	10,948.04 ± 1.43
6	Ethyl isovalerate*	108–64-5	354.29 ± 0.02	390.85 ± 0.02
7	Isoamyl acetate	123–92-2	415.23 ± 0.10	414.36 ± 0.15
8	Ethyl valerate	539–82-2	5614.38 ± 1.13	5998.71 ± 1.11
9	Methyl hexanoate	106–70-7	176.34 ± 0.05	143.22 ± 0.05
10	Butyl butyrate	109–21-7	92.78 ± 0.01	93.61 ± 0.01
11	Ethyl hexanoate	123–66-0	211,566.23 ± 44.92	222,505.26 ± 43.10
12	propyl hexanoate	626–77-7	605.65 ± 0.13	568.72 ± 0.13
13	Ethyl heptanoate	106–30-9	3667.73 ± 0.74	3499.93 ± 0.72
14	Ethyl lactate	97–64-3	138,350.65 ± 30.17	157,860.48 ± 32.83
15	Ethyl caprylate	106–32-1	4110.00 ± 0.72	2897.50 ± 0.52
16	3-Methyl butyl hexanoate	2198-61-0	161.46 ± 0.03	176.01 ± 0.03
17	Ethyl 2-Hydroxy-4-Methylvalerate	10,348–47-7	564.79 ± 0.04	534.52 ± 0.04
18	Ethyl phenylacetate*	101–97-3	–	249.92 ± 0.02
19	Phenethyl phenylacetate	102–20-5	864.87 ± 0.10	887.18 ± 0.07
20	Ethyl laurate	106–33-2	–	119.85 ± 0.01
21	Ethyl phenylpropiolate	2216-94-6	279.04 ± 0.03	232.29 ± 0.03
22	Benzylcarbinyl caproate	6290-37-5	271.73 ± 0.05	293.89 ± 0.04
23	Ethyl palmitate	628–97-7	3298.85 ± 0.18	3224.43 ± 0.18
24	Ethyl Oleate*	111–62-6	1175.00 ± 0.05	1082.50 ± 0.05
25	ethyl linoleate	544–35-4	1841.66 ± 0.17	1889.07 ± 0.12
	Alcohols			
1	Methyl alcohol	67–56-1	17,050.52 ± 0.86	17,933.21 ± 0.72
2	2-butanol	78–92-2	5188.61 ± 1.08	5916.05 ± 1.24
3	1-propanol	71–23-8	45,135.00 ± 4.00	47,255.00 ± 3.98
4	Isobutanol	78–83-1	7322.54 ± 1.15	7178.60 ± 0.98
5	2-Pentanol	6032-29-7	501.18 ± 0.10	504.22 ± 0.11
6	1-Butanol	71–36-3	34,237.50 ± 1.62	33,633.75 ± 3.54
7	Isoamyl alcohol	123–51-3	23,872.41 ± 1.64	23,367.37 ± 1.50
8	pentanol	71–41-0	3554.84 ± 0.18	3324.83 ± 0.28
9	2-Heptanol	543–49-7	207.38 ± 0.03	203.18 ± 0.03
10	1-Hexanol	111–27-3	14,403.75 ± 0.43	14,093.75 ± 1.42
11	2-Pentanol, 3-ethyl-	609–27-8	154.85 ± 0.01	154.30 ± 0.02
12	Meso-2,3-butanediol	5341-95-7	677.44 ± 0.08	697.08 ± 0.07
13	Propylene Glycol	57–55-6	2613.97 ± 0.27	2673.68 ± 0.28
14	2,3-Butanediol	513–85-9	1133.75 ± 0.19	930.00 ± 0.11
15	3-Furancarbinol*	4412-91-3	425.52 ± 0.03	477.09 ± 0.01
16	Phenethyl alcohol	60–12-8	318.84 ± 0.06	326.79 ± 0.06
17	Glycerol	56–81-5	–	111.66 ± 0.01
	Aldehydes			
1	Acetaldehyde*	75–07-0	11,066.25 ± 1.29	7800.00 ± 0.64
2	Acetal*	105–57-7	30,463.75 ± 4.36	20,675.00 ± 3.23
3	Isovaleraldehyde	590–86-3	2803.75 ± 0.41	3433.75 ± 0.57
4	Hexanal	66–25-1	209.65 ± 0.04	273.26 ± 0.04
5	Furfural	98–01-1	2520.91 ± 0.41	2989.92 ± 0.55
6	Benzaldehyde	100–52-7	276.23 ± 0.08	263.11 ± 0.07
7	Phenylacetaldehyde	122–78-1	435.04 ± 0.08	394.13 ± 0.05
	Ketones			
1	Acetone	67–64-1	2049.59 ± 0.51	2173.11 ± 0.39
2	2-butanone	78–93-3	785.00 ± 0.15	948.75 ± 0.18
3	2-Pentanone	107–87-9	201.19 ± 0.06	225.26 ± 0.05
4	2-Heptanone*	110–43-0	342.50 ± 0.05	437.50 ± 0.04
5	Acetoin*	513–86-0	1923.75 ± 0.21	2412.50 ± 0.26+
6	Acetol	116–09-6	186.64 ± 0.02	188.60 ± 0.05
7	3-methyl-2(5H)-furanone	22,122–36-7	314.81 ± 0.07	365.97 ± 0.05
	Acids			
1	Acetic acid	64–19-7	120,907.88 ± 13.76	119,647.69 ± 0.57
2	Propionic acid	79–09-4	2868.01 ± 0.43	2941.15 ± 0.31
3	Isobutyric acid	79–31-2	838.42 ± 0.07	917.22 ± 0.07
4	Butyric Acid	107–92-6	8590.76 ± 0.49	8827.57 ± 0.39
5	3-Methylbutanoic acid	503–74-2	1484.33 ± 0.14	1484.10 ± 0.12
6	Valeric acid	109–52-4	2241.10 ± 0.12	2449.52 ± 0.16
7	1-Hexanoic acid*	142–62-1	34,925.37 ± 1.63	38,744.13 ± 2.33
8	Heptanoic Acid	111–14-8	600.49 ± 0.14	617.33 ± 0.14
9	Octanoic acid	124–07-2	390.65 ± 0.04	380.01 ± 0.04
	Nitrogenous compounds			
1	Trimethyl-pyrazine	14,667–55-1	1092.50 ± 0.25	853.75 ± 0.13
	Other compounds	0		
1	1,1-diethoxy-2-methyl-Butane	3658-94-4	291.69 ± 0.06	293.08 ± 0.08
2	1,1-diethoxy-3-methyl-Butane	3842-03-3	2193.76 ± 0.47	2586.38 ± 0.64
3	1,1-Diethoxyhexane	3658-93-3	215.27 ± 0.05	209.50 ± 0.05

Table 4
Major components with OAV > 1 of Baijiu (* P<0.05).

Code	Aroma compound	Odor threshold (μg/L)	Odor description	OAV LS	OAV FS
1	Ethyl acetate [a]	32,551.6	Pineapple, apple, fruit	5.45 ± 0.69	5.50 ± 6.64
2	Ethyl isobutyrate * [b]	57.47	Apple, pear	7.69 ± 0.63	9.13 ± 0.88
3	Ethyl butyrate [a]	81.5	Pineapple	132.92 ± 17.97	134.33 ± 17.53
4	ethyl isovalerate * [b]	6.89	Strawberry, apple, fruity	51.42 ± 2.42	56.73 ± 3.21
5	Isoamyl acetate [a]	93.93	Banana, sweet, apple	4.42 ± 1.10	4.41 ± 1.64
6	Ethyl valerate [a]	26.78	Yeast, fruity	209.65 ± 42.15	224.00 ± 41.30
7	Ethyl hexanoate [b]	55.33	Apple, pear	3823.72 ± 811.81	4021.42 ± 778.92
8	Ethyl lactate [a]	128,083.8	Sour fruit, creamy	1.08 ± 0.24	1.23 ± 0.26
9	Ethyl caprylate * [b]	12.87	Fruity, greasy	319.35 ± 56.19	225.14 ± 40.04
10	Ethyl palmitate [a]	1500	Waxy, creamy	2.20 ± 0.12	2.15 ± 0.12
1	1-Butanol [b]	2733.35	Yeast, fruit	12.53 ± 0.59	12.30 ± 1.30
2	1-Hexanol [c]	5370	Grass, floral	2.68 ± 0.08	2.62 ± 0.26
1	Acetaldehyde * [d]	1200.03	Floral, grass	9.22 ± 1.08	6.50 ± 0.53
2	Acetal * [d]	2086.5	Grass, fruit	14.60 ± 2.09	9.91 ± 1.55
3	Isovaleraldehyde [e]	16.51	Fruity. floral	169.82 ± 24.55	207.98 ± 34.76
4	Hexanal [f]	25.48	Grass, floral, fruit	8.23 ± 1.42	10.68 ± 1.66
5	Phenylacetaldehyde [c]	261.95	–	1.66 ± 0.30	1.50 ± 0.19
6	Furfural [g]	122	–	20.66 ± 3.38	24.51 ± 4.48
1	2-Pentanone [f]	138	–	1.46 ± 0.40	1.63 ± 0.37
2	Acetoin * [f]	259	Butter	7.43 ± 0.79	9.31 ± 1.02
3	2-Heptanone * [g]	140	–	2.45 ± 0.38	3.13 ± 0.29
1	Butyric Acid [h]	964.64	Smell of sweat, acid, cellar mud	8.91 ± 0.50	9.15 ± 0.40
2	3-Methylbutanoic acid [c]	1045.47	–	1.42 ± 0.13	1.42 ± 0.12
3	Valeric acid [h]	389.11	–	5.76 ± 0.32	6.30 ± 0.40
4	1-Hexanoic acid * [h]	2517.16	Sour, vinegar	13.87 ± 0.65	15.39 ± 0.92
1	Trimethyl-pyrazine [c]	729.86	–	1.50 ± 0.34	1.17 ± 0.18

a Odor thresholds from reference (Du et al., 2023).
b Odor thresholds from reference (Wu et al., 2024).
c Odor thresholds from reference (Yang et al., 2024).
d Odor thresholds from reference (Guo et al., 2019).
e Odor thresholds from reference (Yu et al., 2024).
f Odor thresholds from reference (Ma et al., 2024).
g Odor thresholds from reference (Liu & Sun, 2018).
h Odor thresholds from reference (Wu et al., 2022).

of lipids, fatty acyls (FA), glycerolipids (GL), sphingo lipids (SP), glycerophospholipids (GP), prenol lipids (PR), and sterol lipids (ST) were identified in Baijiu, and 26 lipid subclasses were detected, with a total of 239 lipids identified. A list of all the lipid compounds with average Rt (min), average Mz, metabolite name, formula, ontology, and category is shown in Supplementary Table 2. The lipid percentages of different Baijiu samples are shown in supplementary Fig. 4b. The highest FA content was observed, followed by GL, SP, PR, GP, PR, and ST.

supplementary Fig. 4c and supplementary Fig. 4d show the lipid profiles of Baijiu with 12 acylcarnitines (CAR), 36 n-acyl ethanolamines (NAE), 39 n-acyl glycines (NAGly), 4 n-acyl glycyl serines (NAGlySer), 11 n-acyl ornithines (NAOrn), 15 fatty acid ester of hydroxyl fatty acids (FAHFA), 5 free fatty acids (FA), 20 oxidized fatty acids (OxFA), 13 diacylglycerols (DG), 8 ETHER-linked diacylglycerols (EtherDG), 7 monoacylglycerols (MG), 1 triacylglycerols (TG), 3 ether-linked lysophosphatidylethanolamines (EtherLPE), 4 lysophosphatidylglycerols (LPG), 9 lysophosphatidic acids (LPA), 1 lysophosphatidylethanolamines (LPE), 3 coenzyme Qs (CoQ), 1 vitamin a fatty acid esters (VAE), 4 Ceramide non-hydroxyfatty acid-dihydrosphingosines (Cer_NDS), 4 ceramide hydroxy fatty acid-dihydrosphingosines (Cer_HDS), 3 ceramide alpha-hydroxy fatty acid-phytospingosines (Cer_AP), 11 ceramide non-hydroxyfatty acid-sphingosines (Cer_NS), 2 ceramide hydroxy fatty acid-sphingosines (Cer_HS), 7 phytosphingosines (PhytoSph), 7 sphinganines (DHSph), and 9 sterols (ST). The major lipid components were NAE (FA 35.69 %–39.20 %), CAR (FA 23.76 %–23.92 %), OxFA (FA 13.42 %–16.57 %), MG (GL 5.09 %–6.19 %), DG (GL 4.31 %–6.05 %), NAGly (FA 3.70 %–3.79 %), and the rest of the types of lipid molecules constituted less than 2 %.

3.7. Discrepant analysis of lipids in different baijiu samples

We identified 239 lipid subclasses from 26 categories using a lipidomics approach to differentiate between Baijiu samples derived from various raw material sources of LS and FS. The OPLS-DA score plot revealed that the samples in the LS and FS groups were dispersed, whereas the samples within each group clustered together. This indicated a significant difference in lipid composition between the LS and FS groups (Fig. 3a). Cross-validation using 1000 permutation tests confirmed the reliability of the OPLS-DA model, with Q2 and R2Y values of 0.752 and 0.97, respectively (Fig. 33b).

VIP was used to identify the most significant lipids. Fig. 3c summarizes the important lipids (VIP > 1.0) based on OPLS-DA analysis. A total of 26 lipids can serve as markers to differentiate between the various Baijiu samples. In the discrepant lipids, AAHFA 5:0/8:0;O, AAHFA 6:0/15:4;O, CAR 4:0, FA 14:0, FA 14:2, FA 18:1;O3, FA 18:2;O3, FA 18:3;O3, FA 20:1;O3, FA 20:2;O2, FA 22:6;O4, NAE 12:0, NAE 14:0, NAE 19:1, NAE 21:4, NAE 9:0, and NAGly 22:4, which were classified as FA Additionally, DG 17:0, MG 8:1, and MG 9:0 were classified as GL, whereas LPA 13:1, LPA 15:2, LPG 5:0, and LPG 9:0 were classified as GP, SPB 16:0;O3, SPB 19:0;O2 were classified as SP.

To investigate the potential relationships among the lipids in the analyzed Baijiu samples, we calculated the Pearson correlation coefficients for the 26 lipids using a visualization heatmap (Fig. 3d). Notably, CAR 4:0, MG 9:0, AAHFA 6:0/15:4, FA 18:2;O3, FA 18:1;O3,

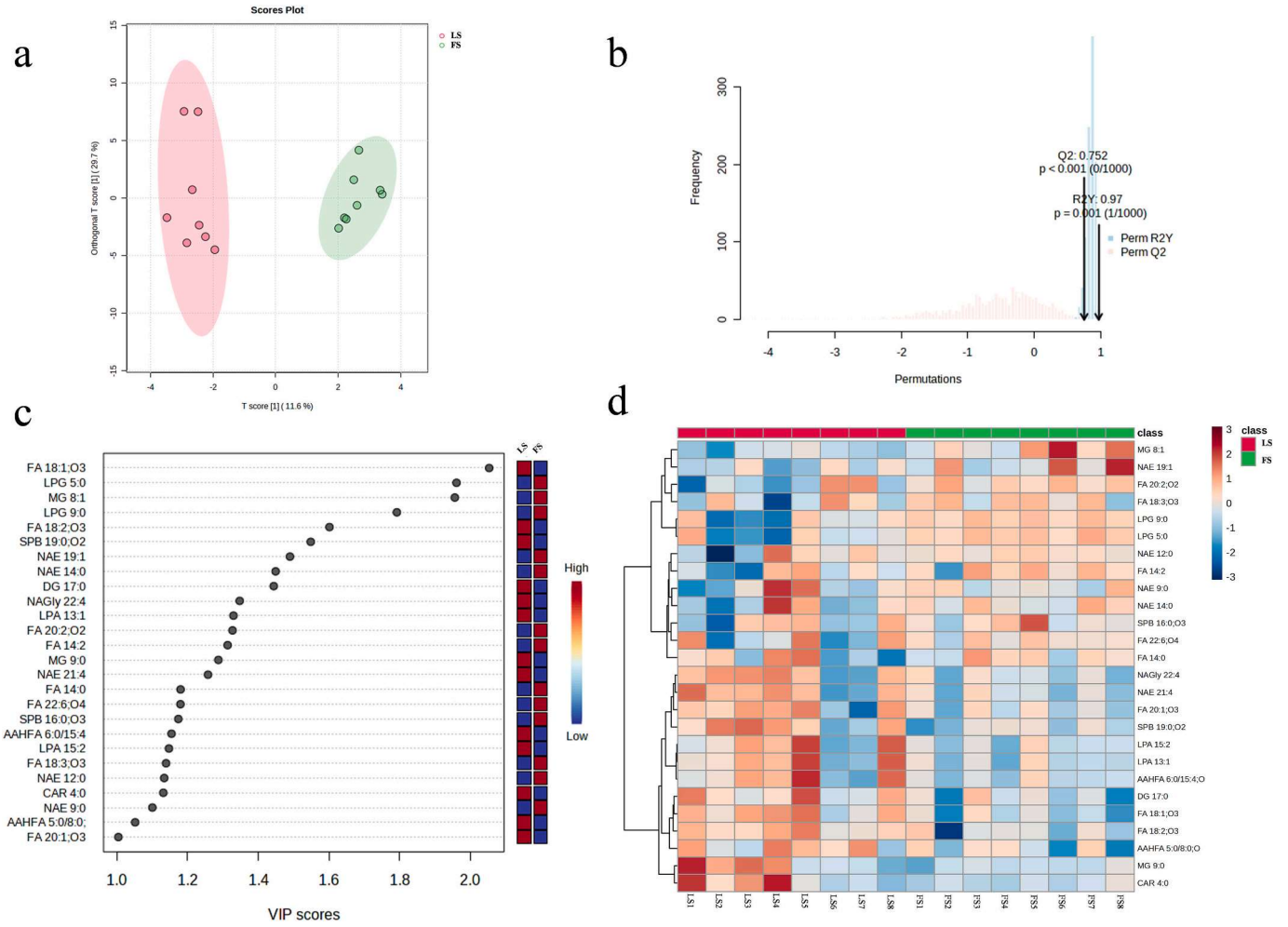


Fig. 3. Results of the lipid profiles of Baijiu made from different sorghums. a). OPLS-DA of sensory evaluation results. b). OPLS-DA model permutation. c). 26 discrepant lipids (VIP>1) of different Baijiu. d). The heatmap of 26 lipids of different Baijiu.

and DG 17:0 exhibited significant positive correlations in the LS group. Similarly, NAE 12:0, FA 14:2, LPG 5:0, LPG 9:0, FA 18:3;O3, FA 20:2;O2, NAE 19:1, and MG 8:1 showed significant positive correlations in the FS group.

A total of 142 lipids were identified in the FA category. The percentage of NAE, CAR, OxFA, NAGly, FAHFA, FA, NAOrn, and NAGlySer in the lipid subclass were 44.01 %–47.97 %, 29.32 %–29.35 %, 16.37 %–20.14 %, 4.53 %–4.67 %, 0.75 %–0.93 %, 0.50 %–0.53 %, 0.28 %–0.31 %, 0.04 %–0.05 %, respectively. A significant difference was observed in the OxFAs. A total of 29 lipids were identified in the GL categories, of which the percentages of MG, DG, EtherDG, and TG contents were 41.62 %–49.45 %, 37.63 %–47.81 %, 10.43 %–12.73 %, 0.144 %–019 %, respectively. Additionally, 17 lipids were identified in GP, of which the contents of LPA, LPD, EtherLPE, LPE were 32.80 %–59.84 %, 23.51 %–53.15 %, and 13.29 %–53.37 %, 0.76 %–1.28 %, respectively, of which LPA and LPG were significantly different between the two samples (Supplementary Fig. 5).

The presence of lipids in the product strongly influences aroma release, and the volatility of the aromatic compounds in the product varies depending on their affinity for food components. The chemical structure of lipids affects the flavor profile of the product. Lipids strongly influence flavor perception, duration, and flavor release (de Roos, 2006). In addition, lipids significantly contribute to the classification of wine origin. There are differences in the lipid compositions of wines made from grapes of different origins (Phan & Tomasino, 2021), suggesting that the differences in lipids found in this study can be used to determine the origin of sorghum and the selection of brewing materials. Different treatments affect the free fatty acid (FFA) (Waterhouse et al., 2016). Besnard (Besnard et al., 2016) and Mattes (Mattes, 2009) both agreed that long-chain fatty acids are an effective taste stimulus, whereas FFA are the main component associated with fatty mouth feeling; specifically, the longer and more saturated fatty acids, which contribute to a smoother, creamier texture (Zhou et al., 2016). The differential substances screened, among which LS contained FA 14:0, FA 16:0, FA 16:1, and FA 22:0, were higher than those in FS. The relative content of FA 14:2 in FS was higher than that in LS, which was hypothesized to be the reason for the differences in taste among the different samples. In addition, phospholipids (PAs) and DGs are associated with wine mouthfeel, providing an oily, soft, and velvety mouthfeel to wines (Waterhouse et al., 2016). In wine-related studies, the addition of PAs alters the mouthfeel of model wines by increasing their perceptions of viscosity. When PA is present, it helps mask bitterness and enhance the perception of acidity (Shen, 1998). A related study found that wine contains numerous phenolic compounds and proteins and that MG and DG can form complexes in aqueous solutions (Loi et al., 2019). TGs interact with polyphenols and alter taste perception (Reis et al., 2020). We hypothesized that GLs would influence wine taste by forming complexes with other chemical components. In addition to directly affecting mouthfeel, lipids may interact with other compounds in wine to alter mouthfeel perception. Thus, altering lipid composition may be a viable option to change mouthfeel quality.

The sources of nonvolatile compounds in Baijiu include raw

materials, the fermentation process, microbial metabolism, and the Baijiu storage process (Huang et al., 2022); a small portion enters Baijiu through distillation, and the vast majority originates from the aging process of Baijiu (Jia et al., 2021; Jia & Ma, 2023; Jiang et al., 2021; Zhang, Li, et al., 2022). Some lipids are closely associated with biological effects and are key to organisms' nutritional and energy sources (Wu, Wei, et al., 2021). These are an important source of nutrients for *Saccharomyces cerevisiae* under harsh anaerobic brewing conditions (Pinu et al., 2014; Tesniere, 2019). Studies have shown that some lipids are closely associated with biological effects. N-Acyl ethanolamines (NAEs) are fatty acids linked to ethanolamine by an amide bond. They are classified based on the number of carbons and the degree of saturation of their acyl chains. Most studies on NAEs have focused on vertebrates; however, NAEs have also been detected in various other organisms, including yeast (*S. cerevisiae*) and *Dictyostelium discoideum*. Peroxidative stress leads to elevated NAE levels in different *S. cerevisiae* species (Merkel et al., 2005). It is hypothesized that the environmental conditions of the yeast during fermentation lead to differences in the NAE content of Zaopei, ultimately leading to differences in the NAE content of Baijiu.

Furthermore, glycerol phospholipids and sphingolipids were found to be key lipids in the response to acetic acid stress. For example, acetic acid stress increases glycerophospholipid saturation in *Zygosaccharomyces bailii*. It also promotes the supply of fatty acids via glycerol phospholipids to synthesize sphingolipids, which enrich and coordinate sphingolipids, thereby improving acetic acid tolerance (Lindberg et al., 2013). The well-established biological function of acylcarnitines (CARs) is to transport acyl groups from the cytosol into the mitochondrial matrix for oxidation, which produces the energy necessary to sustain cellular activity. This process facilitates the transport of long-chain fatty acids to mitochondria (Indiveri et al., 2011; Reuter & Evans, 2012). Lysophosphatidic acid (LPA) is closely associated with biological effects, and LPA with shorter fatty acyl chains exhibits lower biological activity. LPA exhibits maximum biological activity when its fatty acyl chain length is in the C16-C20 range (Jalink et al., 1994; Moolenaar, 1995). This indicates that the lipid sources in Baijiu are closely linked to the metabolic activities of the microorganisms.

As previous findings showed that glutinous sorghum in southern Sichuan Province significantly impacted triglyceride metabolism in Zaopei, it was speculated that lipid composition differences in fermented grains led to lipid composition differences in Baijiu. Therefore, we focused on the lipid profile of Baijiu. Subsequent relevant studies could begin with lipid differences in grains, the effects of different lipid components on the lipid composition of Zaopei, and the lipid spectrum of Baijiu.

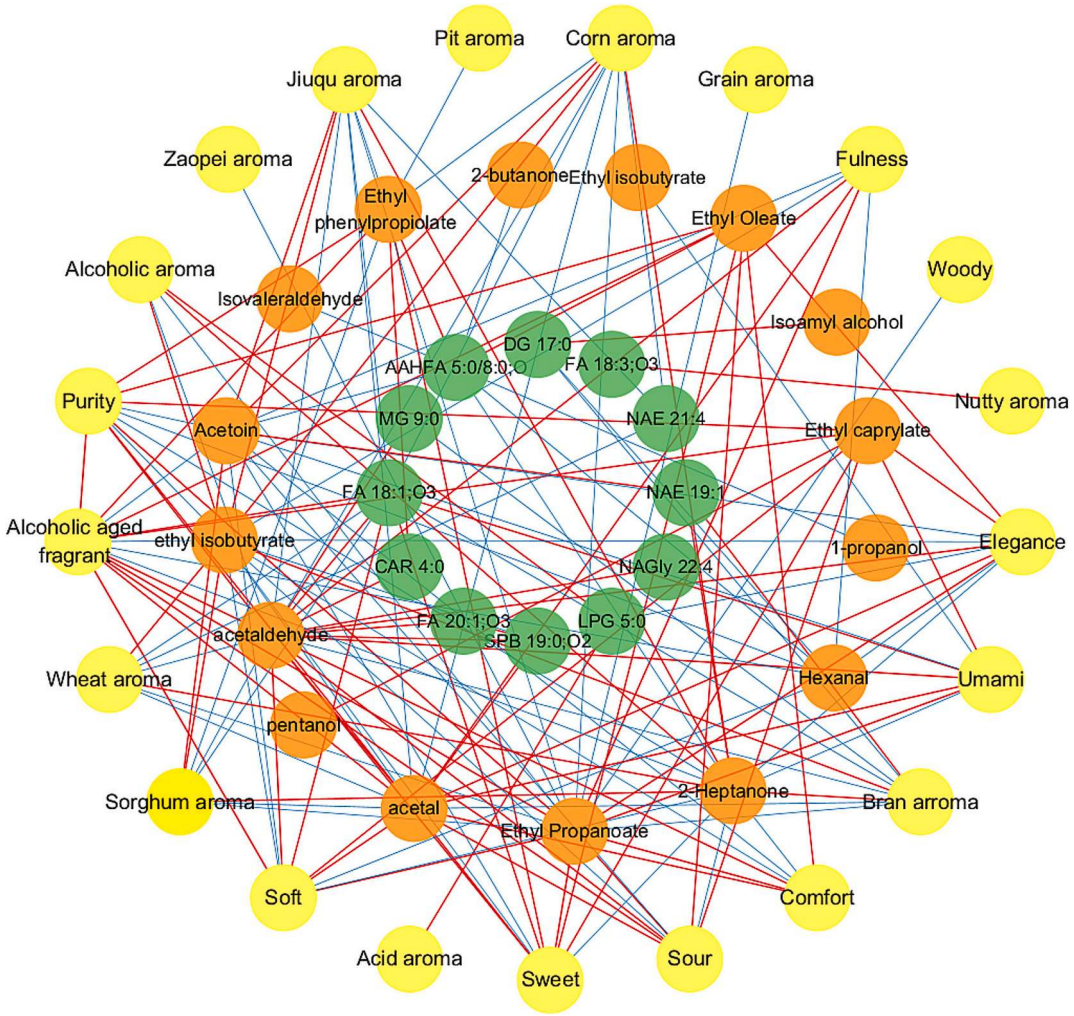


Fig. 4. Correlation between lipids, VOCs, and sensory characterizations. The green circles represent lipids, the orange circles represent VOCs, and the yellow circles represent sensory characterizations. The red line shows a positive correlation, and the blue line shows a negative correlation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.8. Correlation analysis of sensory characteristics, VOCs, and lipids

The flavor of Baijiu depends on the interaction between volatile and non-volatile compounds. Certain nonvolatile compounds are taste-active substances that may interact with VOCs, thus affecting the volatility of aromatic compounds (Li et al., 2020; Niu et al., 2019). The interaction between VOCs and nonvolatile compounds can result in either synergistic or masking effects on flavor, which are determined by their distinct chemical structures and molecular polarities, ultimately impacting sensory evaluation. Additionally, VOCs can modulate the taste properties of nonvolatile compounds, thereby harmonizing the overall flavor profile of Baijiu (Duan et al., 2022). Although lipids are not abundant in Baijiu, their interactions with other compounds may significantly affect its flavor. To explore the potential relationship between lipids, VOCs, and sensory characterizations, we selected 21 sensory discrepancy indicators, 20 GC–MS discrepancy substances, and 26 lipid substances (VIP $>$ 1) for correlation analysis, and their network correlation diagrams are shown in Fig. 4. (Spearman $>$0.6). The results showed that acetaldehyde and ethyl caprylate were positively correlated with aged fragrant (alcoholic aged fragrant), aroma sensation (elegance and comfort), and taste (sour, sweet, and umami). FA 18:1;O3 also positively correlated with aged fragrance, taste, and mouthfeel.

GC–MS was used to identify VOCs in Baijiu samples from different years, and compounds such as ethyl caproate, acetaldehyde, and ethyl caprylate were screened for their significant correlation with Baijiu's storage age (Li, Yuan, et al., 2020). The volatile profiles of young and aged Qingxiangxing Baijiu were characterized, and acetaldehyde was identified as a distinguishing compound in both (Sun et al., 2023); the content of acetaldehyde in Baijiu showed a rising trend during storage and aging, and it was hypothesized that the aroma of Baijiu aging was related to acetaldehyde. A significant difference in acetaldehyde was observed in the GC–MS analysis. Researchers combined sensomics and flavoromics to find that the contributors to the sweetness of Baijiu were associated with compounds such as ethyl caprylate, ethyl heptanoate, and ethyl hexanoate, and ethyl caprylate also showed a significant positive correlation with the sweetness of Baijiu (Sun et al., 2021). FFAs, which are primarily responsible for the perception of fat in the mouth (Martin et al., 2011), contribute increasingly to a smooth mouthfeel as their chain length and saturation increase (Zhou et al., 2016). Long-chain fatty acids (oleic acid) inhibit the volatilization rate of volatiles, thus altering the flavor of Baijiu (Q. R. Liu et al., 2024). The correlation network diagram also showed that FA 18:1;O3 were negatively correlated with Baijiu raw material aroma (Grain aroma, Sorghum aroma, et al.). The FA 18:1;O3 relative content of Baijiu LS was higher than that of Baijiu FS, and the sensory evaluation results of Baijiu LS raw material aroma were lower than those of Baijiu FS, which showed a trend consistent with the correlation results presented in the network diagram.

4. Conclusion

In this study, sensory evaluation and, *E*-nose, E-tongue, HS-SPME-GC–MS, and UPLC-TOF-MS/MS techniques were employed to measure and characterize the flavor differences and lipid compositions of SFB produced from different sorghum raw materials. Meanwhile, the influence of lipids on the flavor of Baijiu was explored. There were flavor differences between the two Baijiu samples. In addition, the OAVs of 26 aroma compounds were relatively high, and these compounds were identified as important aroma compounds. They showed differences between the two types of Baijiu and were considered to be the reasons for the flavor differences between the two kinds of Baijiu. The lipid composition affects the flavor and mouthfeel of Baijiu. FA 18:1;O3, acetaldehyde, and ethyl caprylate jointly contributed to the smooth and soft flavor of Baijiu. These findings can be used as a guide for the sselection of brewing materials to improve the quality of Baijiu and the influence of lipid on Baijiu flavor.

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CRediT authorship contribution statement

Hui Zhu: Writing – review & editing, Data curation. **Linjuan Li:** Writing – original draft, Visualization, Software, Conceptualization. **Yi Ma:** Supervision, Investigation. **Huibo Luo:** Supervision. **Ahmed H. El-Sappah:** Writing – review & editing, Formal analysis. **Ying Liu:** Methodology. **Xiaoke Sun:** Visualization. **Shijiang Pan:** Software. **Muhammad Aamer Mehmood:** Writing – review & editing, Formal analysis. **Yong Sun:** Software, Methodology. **Ning Wang:** Writing – review & editing, Validation, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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